

d) Some student has solved every exercise in this book

$S(x,y)$ : Student  $x$  has solved exercise  $y$  in this book

$\exists x \forall y S(x,y)$ .

Domain of  $x$ : All students of the class

Domain of  $y$ : All exercise problems of the book

$$\neg \exists x \forall y S(x,y) \equiv \forall x \neg \forall y S(x,y) \text{ (De Morgan's law of quantifiers)}$$

$$\equiv \forall x \exists y \neg S(x,y) \text{ (De Morgan's law of quantifiers)}$$

For each student, there exists some exercise which has not been solved.

Simple: No student has solved every exercise in this book.

e) No student has solved at least one exercise in every section of this book.

~~$S(x,y,z)$~~   $S(x,y,z)$ : Student  $x$  has solved exercise  $y$  from section  $z$  of this book.

Domain of  $x$ : All students of the school

Domain of  $y$ : All exercise problems of the book

Domain of  $z$ : All sections of the book

$$\forall x \forall z \exists y \neg S(x,y,z)$$

$$\neg \forall x \forall z \exists y \neg S(x,y,z) \equiv \exists x \neg \forall z \exists y \neg S(x,y,z) \text{ (De Morgan's law of quantifiers)}$$

$$\equiv \exists x \exists z \neg \exists y \neg S(x,y,z) \text{ (De Morgan's law of quantifiers)}$$

$$\equiv \exists x \exists z \forall y \neg (\neg S(x,y,z)) \text{ (De Morgan's law of quantifiers)}$$

$$\equiv \exists x \exists z \forall y S(x,y,z) \text{ (Double-negation law)}$$

There exists some student who has solved all exercises from some section of this book.

37) Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English (Do not simply use the phrase, "It is not the case that.")

a) Every student in this class has taken exactly two mathematics classes at this school.  $T(x,y,z)$ : Student  $x$  has taken class  $y$  from department  $z$